

Functions In Calculus

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1 Introduction

To “compile” this information I am using the book Calculus by David Patrick it can be found here: <https://artofproblemsolving.com/store/book/calculus>.

This Document is designed to review the Function principals covered in the aforementioned book. I will create other “Guides” for other topics covered. They can be found here: <https://github.com/KovasMcCann/Writings>.

2 What is a Function

A function is a matamatical device that associates an input with a unique output.

2.1 Example

$$f(x) = 3x - 5$$

Input: $x \in \mathbb{R}$

Output: $3x - 5 \in \mathbb{R}$

2.2 Formal Definition

A function f from set A to set B

$$f : A \rightarrow B$$

Where:

$$a \in A \text{ and } f(a) \in B$$

3 Properties of Functions

There are three main properties of functions: **Domain**, **Codomain**, and **Range**.

- The **Domain** is the set of all possible input values of a function. It is denoted by $Dom(f)$.
- The **Codomain** is the set of all potential output values defined for the function, whether or not they are actually produced. It is denoted by $Cod(f)$.
- The **Range** is the set of all actual output values that the function produces from the domain. It is denoted by $Rng(f)$.

3.1 Example

Let $f(x) = x^2$, where $x \in \mathbb{R}$.

- $Dom(f) = \mathbb{R}$
- $Cod(f) = \mathbb{R}$
- $Rng(f) = [0, \infty)$

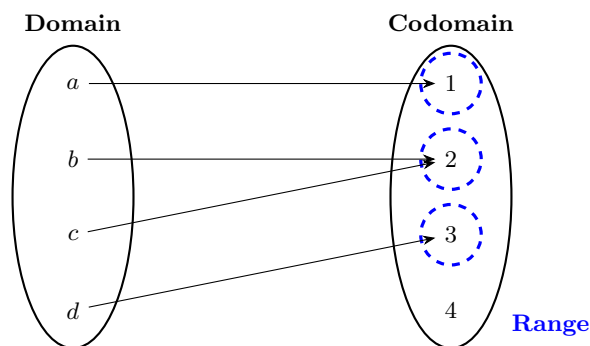


Figure 1: The relationship between input and output in a function. The first set represents the domain (inputs), the second set represents the codomain (all possible outputs), and the blue dashed circles highlight the actual outputs — the range.

4 Operations

All basic arithmetic operations can be applied to functions themselves.

Given two functions $f(x)$ and $g(x)$, we can define the following operations:

- **Addition:**

$$(f + g)(x) = f(x) + g(x)$$

The resulting function represents the pointwise addition of f and g .

- **Subtraction:**

$$(f - g)(x) = f(x) - g(x)$$

This represents the pointwise subtraction of g from f .

- **Multiplication:**

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

The multiplication is also performed pointwise.

- **Division:**

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad \text{provided that } g(x) \neq 0$$

Division is defined only at points where $g(x) \neq 0$.

- **Scalar Multiplication:**

$$(c \cdot f)(x) = c \cdot f(x)$$

where c is a constant. This scales the output of the function by c .

- **Composition:**

$$(f \circ g)(x) = f(g(x))$$

This represents function composition, where the output of g becomes the input to f .

5 Inverse Functions

The inverse of a function is where the X and Y values are swapped. The inverse of a function is denoted by f^{-1}

Example:

$$f(x) = 3x - 5$$

$$f^{-1}(x) = \frac{x + 5}{3}$$

5.1 How to Find the Inverse of a Function

To find the inverse of a function, follow these steps:

- Replace each occurrence of $f(x)$ with y
- Swap x and y in the equation
- Solve the resulting equation for y
- Replace y with $f^{-1}(x)$

Example: Find the inverse of $f(x) = 5x - 2$

$$\begin{aligned}y &= 5x - 2 \\x &= 5y - 2 && \text{(Swap } x \text{ and } y) \\x + 2 &= 5y && \text{(Add 2 to both sides)} \\\frac{x + 2}{5} &= y && \text{(Divide both sides by 5)} \\f^{-1}(x) &= \frac{x + 2}{5}\end{aligned}$$

6 Graphs of Functions

Function's are typically graphed on the **Cartesian plane**.

The Cartesian plane is the set of all ordered pairs of real numbers; i.e:

$$\{(x, y) : x, y \in \mathbb{R}\}$$

Normally its depicted either like $\mathbb{R} \times \mathbb{R}$ or $\mathbb{R}^2$.

7 Functions Vs. Equations

A function has at least 2 variables: an output variable and one or more input variables. An equation states that two expressions are equal, and it may involve any number of variables (none, one, or more). A function can often be written as an equation, but not every equation is a function.

This is because functions use the X and Y planes as input and output where $X(Dom())$ is the input and $Y(Cod())$ is the output

Keep this in mind while thinking about Functions.

8 Trigonometric Functions

9 Exponentials and Logarithms

9.1 The relation ship between Exponentials and Logarithms